

WASP-52b: EXHALE Atmospheric-Escape Model vs. He 10830 and H α Observations

Abstract

We model the upper atmosphere and escaping wind of the hot Jupiter WASP-52b with EXHALE (the extended ATES code) and compare the synthetic He I 10830 and H α transit transmission spectra against published observations. The setup follows Yan et al. (2022): the stellar XUV/FUV spectral energy distribution is taken from ϵ Eri (MUSCLES) scaled to the planet’s orbit, the lower boundary is pressure-anchored at 1 μ bar, and the full Roche (tidal) potential is used. The steady-state mass-loss rates reproduce Yan et al. (2022) to within $\sim 10\%$ at every XUV level. The He 10830 absorption is shown to be governed by the metastable He 2^3S population: with the physically motivated L_1 -truncated domain, a He-poor composition (H/He $\simeq 98/2$) and the best-fit XUV flux ($0.5 F_0$), the model reproduces the observed He 10830 line depth ($\sim 4\%$ vs. $\sim 3.4\%$). An extended $10 R_p$ domain over-predicts the absorption by 3–10 $\times$ because of an unphysical He 2^3S tail beyond L_1 . The H α prediction is not trusted because the Ly α radiative transfer that sets the $n = 2$ hydrogen population is not solved here.

1. Target and goal

WASP-52b is a low-density hot Jupiter on a 1.75 d orbit around an active K2V star; excess absorption has been detected in both He 10830 (Kirk et al. 2022) and H α (Chen et al. 2020). Our goal is to compute a *base* EXHALE model (no fitting) and overplot the synthetic He 10830 and H α transmission spectra on the observations. The adopted planet/star parameters (Table 1) match Yan et al. (2022), whose hydrodynamic code (Yang & Guo 2018) we use as the physics reference.

Table 1: Adopted WASP-52 / WASP-52b parameters (Yan et al. 2022).

Quantity	Value	Note
Stellar mass $M_\star$	$0.87 M_\odot$	K2V
Stellar radius $R_\star$	$0.79 R_\odot$	
Stellar T_{eff}	5000 K	for the Balmer continuum
Planet mass M_p	$0.46 M_J$	
Planet radius R_p	$1.27 R_J$	base radius
Equilibrium temperature T_0	1304 K	= base temperature
Orbital distance a	0.0272 au	
He/H number ratio	0.087 (solar) ... 0.005	scanned (§7.)
Base pressure	1 μ bar	$\rightarrow n_0 = 5.56 \times 10^{12} \text{ cm}^{-3}$

2. Stellar input spectrum

WASP-52's XUV spectrum is unobserved, so following Yan et al. (2022) we use the ϵ Eri spectrum from the MUSCLES Treasury Survey (France et al. 2016), scaled from Earth to the planet's orbit by $(d_{\epsilon\text{Eri}}/a)^2$ with $d_{\epsilon\text{Eri}} = 3.212$ pc. The scaling reproduces Yan's fiducial values:

Band	This work	Yan et al. (2022)
$F_{\text{XUV}} (1\text{--}912 \text{ \AA})$	$3.44 \times 10^4 \text{ erg cm}^{-2} \text{ s}^{-1}$	3.42×10^4
$\beta_m = F(1\text{--}100)/F(1\text{--}912)$	0.218	0.22
FUV/NUV (912–2600 Å)	$2.63 \times 10^5 \text{ erg cm}^{-2} \text{ s}^{-1}$	$\sim 2.6 \times 10^5$

The FUV/NUV band photoionizes the He 2^3S metastable level and is therefore essential for the He 10830 prediction. EXHALE reads the two-column (wavelength, flux) SED file directly and integrates it for the band luminosities and the frequency-dependent photoionization/heating rates.

Flux normalization (important). Yan et al. divide the substellar F_{XUV} by 4 to account for redistribution over the sphere, defining $F_0 \equiv F_{\text{XUV}}/4 = 8542 \text{ erg cm}^{-2} \text{ s}^{-1}$, and explore $0.25\text{--}8F_0$. In EXHALE the SED is the full substellar flux and the 2-D approximation “Rate/2 + Mdot/2” multiplies it by 0.5 inside the simulation. Hence our $0.25\times$, $0.5\times$, $1.0\times$ fiducial SED map to Yan's $0.5F_0$, $1F_0$, $2F_0$ respectively. Yan's best fit to the observations is $0.5F_0$, i.e. our $0.25\times$ SED.

3. Numerical setup

The lower boundary is pressure-anchored at $1 \mu\text{bar}$ (Base BC: pressure 1.0), giving $n_0 = P/(k_B T_0) = 5.56 \times 10^{12} \text{ cm}^{-3}$, identical to Yan's base condition. The He 2^3S (metastable) network is enabled (Include He23S? True); trace metals (C/N/O+Mg/Ca/Na/Fe) are optionally added via metals.inp. Convergence uses the two-stage PLM→WENO3 march with a du-keyed hand-off to the Newton/JFNK solver. The automatic initial-condition selector (IC mode: auto) chooses a transonic wind seed (the interior cold sonic point sits at $r_c \simeq 2.42 R_p$).

Domain: tidal potential, two outer boundaries. The full Roche potential along the substellar axis is used in all runs. We compare two outer boundaries:

- **Extended ($10R_p$):** matching Yan's domain extent. This required a one-line extension so that an explicit Outer radius is honored in Roche mode while keeping the tidal terms (documented in the user manual / update log).
- **L_1 -truncated ($\simeq 2.53 R_p$):** the default Roche behavior, truncating at the inner Lagrange point where the effective gravity vanishes.

The tidal force is treated identically to Yan/Yang: the momentum equation carries gravity + stellar tide + centrifugal terms. The escape-energy drop from R_p to L_1 in the full Roche form (0.431 in GM_p/R_p units) agrees with the Erkaev et al. (2007) reduced form $K(\lambda) = 1 - 3/2\lambda + 1/2\lambda^3 = 0.428$ to 0.6% for WASP-52b, so the two tidal prescriptions are effectively equivalent here.

4. Mass-loss validation

At matched XUV flux the EXHALE steady-state mass-loss rates reproduce Yan et al. (2022, their Table 4) to within $\sim 10\%$ (Table 2). This confirms that the wind hydrodynamics and energy balance are consistent with the reference code; differences in the line predictions below are therefore not driven by the mass loss.

Table 2: Steady-state mass-loss rate $\log_{10} \dot{M}$ [g s^{-1}] vs. Yan et al. (2022) Table 4 ($\beta_m = 0.22$). Our SED multiple maps to Yan’s F_0 multiple as noted.

Composition	Flux	EXHALE	Yan (2022)
H/He = 92/8	$1 F_0$ (our $0.5\times$)	11.69	11.70
H/He = 92/8	$2 F_0$ (our $1.0\times$)	11.93	11.96
H/He = 98/2	$1 F_0$ (our $0.5\times$)	11.74	11.69
H/He = 98/2	$2 F_0$ (our $1.0\times$)	11.98	11.95
H/He = 92/8	$0.5 F_0$ (our $0.25\times$)	11.47	11.44
H/He = 98/2	$0.5 F_0$ (our $0.25\times$)	11.49	11.41

5. He 10830: extended domain over-predicts

With the $10R_p$ extended domain the model He 10830 transit depth is 15 to 50%, far deeper than the observed $\sim 3.4\%$ (Fig. 1, left). The H α panel is shown but is not trusted (see §8.).

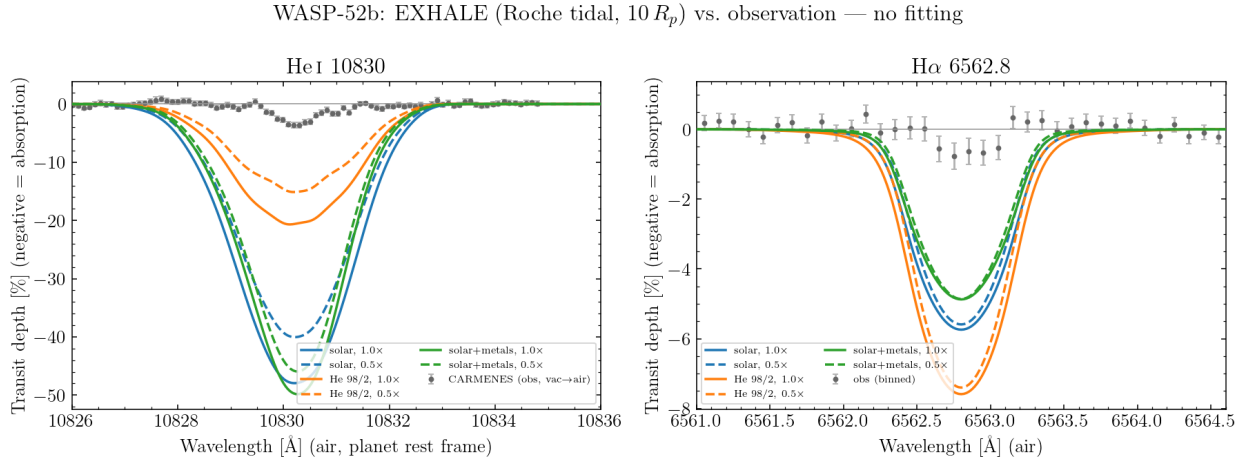


Figure 1: Extended-domain ($10R_p$) model vs. observations. Both He 10830 (left; observation converted vacuum $\rightarrow$ air) and H α (right) are plotted as *transit depth* (negative dip=absorption). Color = composition, line style = XUV level. The model over-predicts He 10830 by 3–10 $\times$.

Diagnosis: the metastable He 2^3S tail. The over-prediction is *not* a mass-loss or flux-normalization error (the $\dot{M}$ matches Yan, §4.). It arises from the spatial distribution of the He 2^3S population. In the steady state

$$\frac{n(\text{He } 2^3S)}{n_{\text{He}^+}} \simeq \frac{\alpha_3 n_e}{\Phi_3 + A_{31} + (q_{31a} + q_{31b})n_e + Q_{31}n_{\text{H}}}, \quad (1)$$

where α_3 is the He $^+$ recombination into 2^3S , Φ_3 the FUV photoionization rate, $A_{31} = 1.27 \times 10^{-4} \text{ s}^{-1}$ the radiative decay, and q_{31}, Q_{31} the electron / hydrogen collisional sinks. We verify $\Phi_3 = 0.061 \text{ s}^{-1}$ (at $1 F_0$) directly from the SED and the NIST cross section, consistent with the FUV flux above. In the cool, rarefied outer wind the collisional sinks vanish, leaving photoionization as the only sink; the ratio $n(\text{He } 2^3S)/n_{\text{He}^+} = \alpha_3 n_e / \Phi_3$ then *rises* outward and the He 2^3S density stays nearly flat from 2 to $5 R_p$. Because the transit depth is weighted by the impact parameter ($\propto r^2$), this sustained outer tail dominates: **72% of the He absorbing column comes from $r > L_1$ ($2.53 R_p$)**. Beyond L_1 the gas is Roche-unbound and physically funnels through L_1 rather than forming a complete spherical shell, so the 1-D spherical treatment to $10 R_p$ over-counts the absorption.

6. He 10830: L_1 -truncated domain reproduces the observation

Truncating the domain at L_1 removes the unphysical outer tail and brings the He 10830 depth into agreement with the data (Table 3, Fig. 2). At the best-fit flux ($0.5 F_0$) and a He-poor composition (H/He=98/2), the model gives 3.9% vs. the observed $\sim 3.4\%$. The solar (92/8) composition remains too strong (7.4%), independently confirming Yan's conclusion that a high H/He ratio is required to match the He 10830 line.

Table 3: He 10830 line-center transit depth [%] (negative = absorption) for the extended ($10 R_p$) and L_1 -truncated domains. Observed $\sim 3.4\%$.

Flux	Composition	$10 R_p$	L_1 -trunc.
$0.5 F_0$	92/8 (solar)	30.6	7.4
$0.5 F_0$	98/2 (He-poor)	11.2	3.9
$1 F_0$	98/2	15.1	4.3
$2 F_0$	98/2	20.7	5.1

7. H/He scan

To locate the composition that matches the He 10830 line we scan the He/H number ratio from solar (92/8) to strongly He-poor (99.5/0.5) at the best-fit flux ($0.5 F_0$) in the L_1 -truncated domain (Table 4, Fig. 4). The He 10830 transit depth shallows (becomes less negative) with decreasing He/H, from 7.4% (solar) to 1.5% (99.5/0.5). The observed depth ($\sim 3.4\%$) is bracketed by H/He = 98/2 (3.9%) and 99/1 (2.5%), implying a best-fit ratio of H/He $\approx 98.3/1.7$ (He/H ≈ 0.017). This is a strongly He-depleted atmosphere relative to solar, in line with the high H/He ratio that Yan et al. (2022) required to reproduce the He 10830 line.

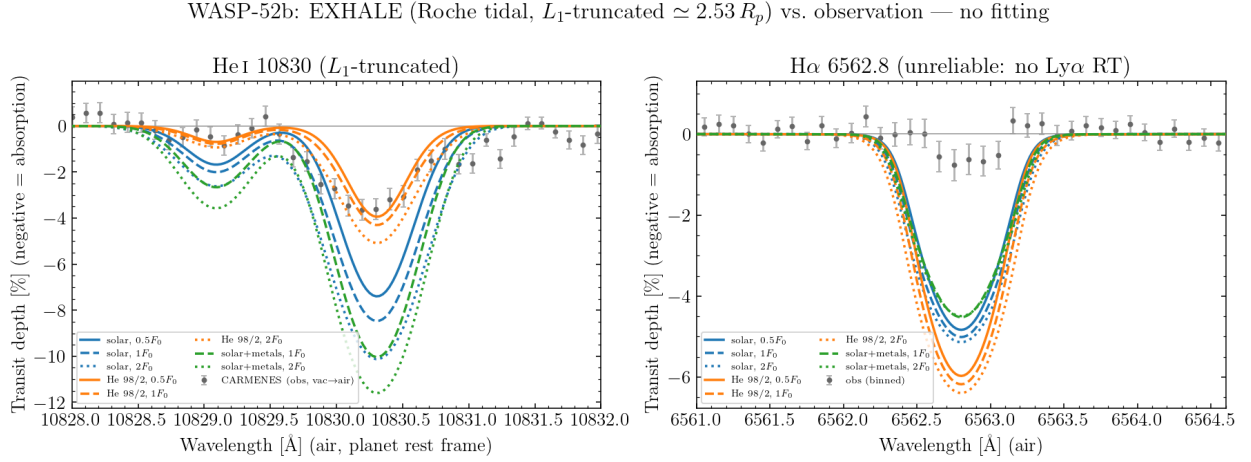


Figure 2: L_1 -truncated ($\simeq 2.53 R_p$) model vs. observations, same layout as Fig. 1. The He-poor $0.5 F_0$ model (orange solid) matches the observed He 10830 line. H α (right) is shown for completeness only.

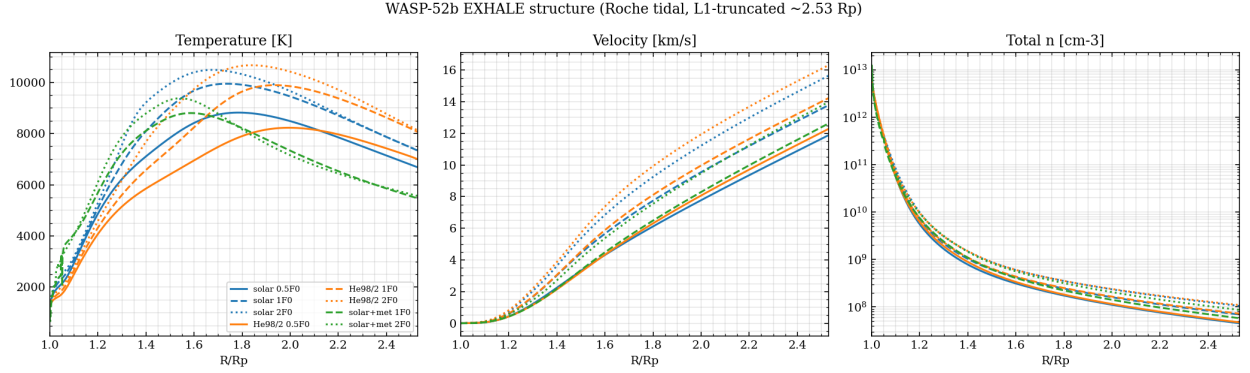


Figure 3: Converged atmospheric structure (temperature, velocity, total number density) for the L_1 -truncated runs. Color = composition, line style = XUV level.

8. H α with a proper Ly α radiative transfer (LaRT)

The H α line absorbs from the $n = 2$ hydrogen population, set primarily by Ly α radiative pumping ($B_{1s \rightarrow 2p} \bar{J}_{\text{Ly}\alpha}$). The simple Huang et al. (2017) estimate of $\bar{J}_{\text{Ly}\alpha}$ used by the bare TPM.py is not adequate, so we instead compute the resonant Ly α radiative transfer with the Monte Carlo code **LaRT** (Seon & Kim 2020) in its spherical-illumination geometry, exactly as in Yan et al. (2022).

Why a 2-D treatment is required. Under stellar (one-sided) illumination the Ly α scattering rate $P_\alpha \equiv B_{1s \rightarrow 2p} \bar{J}_{\text{Ly}\alpha}$ – and hence n_{2p} – is only *cylindrically* symmetric about the star–planet axis, $P_\alpha(\rho, z)$, not spherically symmetric. The spherical TPM.py is therefore replaced for H α by the 2-D integrator `examples/tpm_halpha_lart2d.py`, which reads the LaRT cylindrical scattering-rate grid (Pa_2D), builds $n_{2p}(\rho, z)$ from the Christie et al. (2013) rate equilibrium, and integrates the H α optical depth along the transit line of sight (the star–planet axis).

Table 4: He 10830 line-center transit depth [%] (negative = absorption) vs. composition, at $0.5 F_0$ in the L_1 -truncated domain. Observed $\sim 3.4\%$.

He/H	H/He	He 10830 [%]
0.087	92/8 (solar)	7.40
0.050	95.2/4.8	6.04
0.020	98/2	3.94
0.010	99/1	2.52
0.005	99.5/0.5	1.47

Normalization. LaRT outputs are per unit luminosity, so

$$P_\alpha(\rho, z) = \text{Pa}_{\text{LaRT}}(\rho, z) \times L_{\text{Ly}\alpha} \times \text{fluxfac}, \quad (2)$$

where $L_{\text{Ly}\alpha} = F_{\text{Ly}\alpha} 4\pi a^2 / (h\nu_{\text{Ly}\alpha})$ is the stellar Ly α *photon* luminosity and $\text{fluxfac} = \Omega_\star / 4\pi$ is the geometric factor written to the LaRT header (it is *not* included in the raw scattering-rate accumulation, which uses the photon weight only).

 Table 5: Model parameters for the LaRT-coupled H α calculation.

Quantity	Value	Note
Composition H/He	98/2 ($n_{\text{He}}/n_{\text{H}} = 0.020$)	He-poor (He-fit)
XUV flux F_{XUV}	$0.5 F_0$ ($= 0.25 \times$ fiducial SED)	Yan best fit
$\beta_m = F(1-100)/F(1-912)$	0.218	$\approx$ Yan 0.22
Incident Ly α flux $F_{\text{Ly}\alpha}$	$3.73 \times 10^4 \text{ erg cm}^{-2} \text{ s}^{-1}$	Yan (EP Eri)
Ly α line profile	double-Gaussian, ± 74 , $\sigma = 49 \text{ km s}^{-1}$	Huang 2017
Ly α photon luminosity $L_{\text{Ly}\alpha}$	$4.75 \times 10^{39} \text{ photons s}^{-1}$	$F_{\text{Ly}\alpha} 4\pi a^2 / h\nu$
LaRT photons	10^7 , grid 201 ³	geometry_JPa=2 (cylindrical)

Results. Table 6 and Fig. 6 compare the extended ($10 R_p$) and L_1 -truncated domains. The resulting scattering rates ($P_\alpha \sim 60\text{--}90 \text{ s}^{-1}$) and $n_{2p} \sim (2\text{--}4) \times 10^4 \text{ cm}^{-3}$ are physically reasonable (Christie et al. 2013). Unlike He 10830, the H α depth is *insensitive* to the outer boundary (1.46% at $10 R_p$ vs. 1.28% at L_1): the n_{2p} layer is concentrated near the planet ($r \lesssim 2 R_p$), so the outer tail beyond L_1 contributes little. Both reproduce the line-center bump in the data (~ 0.7 to 1%), with a residual $\sim 1.5\text{--}2 \times$ over-prediction (the model dip is deeper).

The incident Ly α width 49 km/s can be read as the Gaussian σ or as the FWHM (Fig. 5). For the L_1 (Roche) model the narrower FWHM profile gives a *slightly deeper* H α (1.50% vs. 1.28% for $\sigma = 49$), counter to the naive expectation that a narrower pump reduces the absorption: although its peak n_{2p} is lower, the spatial/frequency redistribution of the scattered Ly α shifts the pumping into the layers that dominate the column. The line-profile interpretation is thus a secondary ($\sim 17\%$) effect on H α , much smaller than the $F_{\text{Ly}\alpha}$ lever.

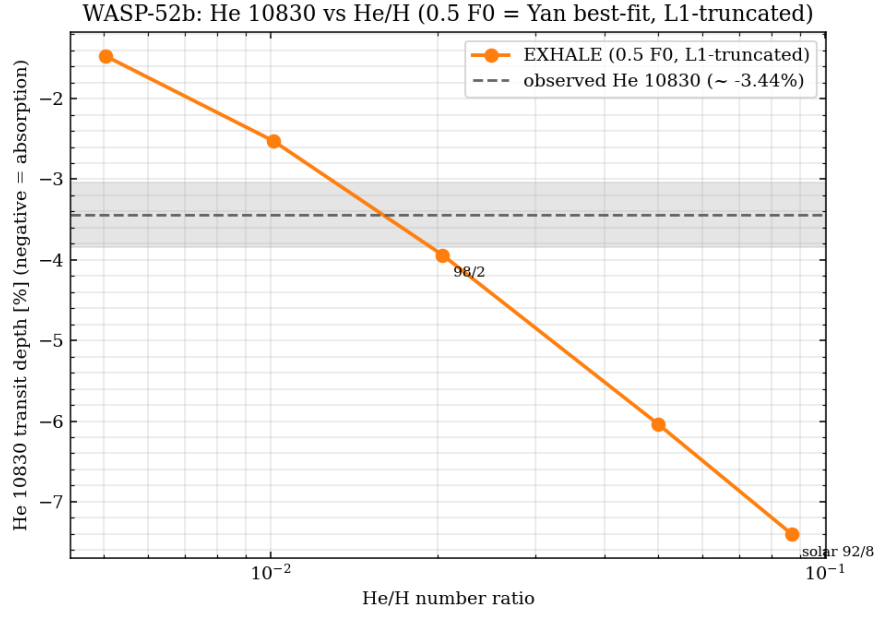


Figure 4: He 10830 line-center transit depth (negative = absorption) vs. He/H number ratio ($0.5 F_0$, L_1 -truncated). The dashed line and gray band are the observed depth ($3.44 \pm 0.4\%$). The model crosses the observation near H/He $\approx 98/2$, confirming the need for a He-depleted atmosphere.

Table 6: LaRT-coupled H α line-center transit depth (negative = absorption) and key quantities (H/He=98/2, $0.5 F_0$, $\beta_m = 0.218$). Ly α width read as σ or FWHM (49 km/s). Observed ~ 0.7 to 1% .

Domain	Ly α width	depth [%]	$P_\alpha^{\max}$ [s $^{-1}$]	$n_{2p}^{\max}$ [cm $^{-3}$]	fluxfac
$10 R_p$ (Yan-like)	$\sigma = 49$	1.46	60.4	2.15×10^4	1.21×10^{-2}
L_1 -trunc. ($2.53 R_p$)	$\sigma = 49$	1.28	85.8	4.19×10^4	7.66×10^{-4}
L_1 -trunc. ($2.53 R_p$)	FWHM= 49	1.50	82.3	1.79×10^4	7.66×10^{-4}

The incident Ly α flux as the H α lever. Because H α is Ly α -pump-dominated, its depth scales with the incident Ly α flux. Rescaling $F_{\text{Ly}\alpha}$ requires *no new LaRT run*: $P_\alpha = P_a \times L_{\text{Ly}\alpha} \times \text{fluxfac}$, so only the luminosity normalization is multiplied. We dropped the $n = 2$ Balmer-continuum photoionization (Γ_2) as a possible lever because it is negligible for this cool host ($\Gamma_2 \simeq 42 \text{ s}^{-1}$ at $T_\star = 5000 \text{ K}$, i.e. $\sim 10^{-7} A_{2p \rightarrow 1s}$). The scan (Fig. 7, Table 7) shows the H α depth shallowing toward zero as $F_{\text{Ly}\alpha}$ decreases (roughly linear; a recombination floor at very low flux). Yan’s $F_{\text{Ly}\alpha} = 37,300 \text{ erg cm}^{-2} \text{ s}^{-1}$ is an EP Eri proxy (WASP-52’s Ly α is unobserved); the observed H α depth is matched at $F_{\text{Ly}\alpha} \approx 0.3\text{--}0.5 \times$ that value for the L_1 (Roche) model ($0.2\text{--}0.3 \times$ for $10 R_p$), i.e. the H α line constrains the effective stellar Ly α flux.

Why the Ly α flux is *not* reduced by the 3-D $\rightarrow$ 1-D factor. The XUV ionizing flux is halved in the hydrodynamics (Rate/2 + Mdot/2, §3.) as a day-side $\rightarrow$ sphere redistribution *approximation*, because the 1-D model is spherically symmetric and cannot resolve the illumination geometry. The incident Ly α flux is *not* reduced by this factor: LaRT’s spherical-illumination run is a genuine 3-D calculation that models the one-sided stellar irradiation explicitly (yielding

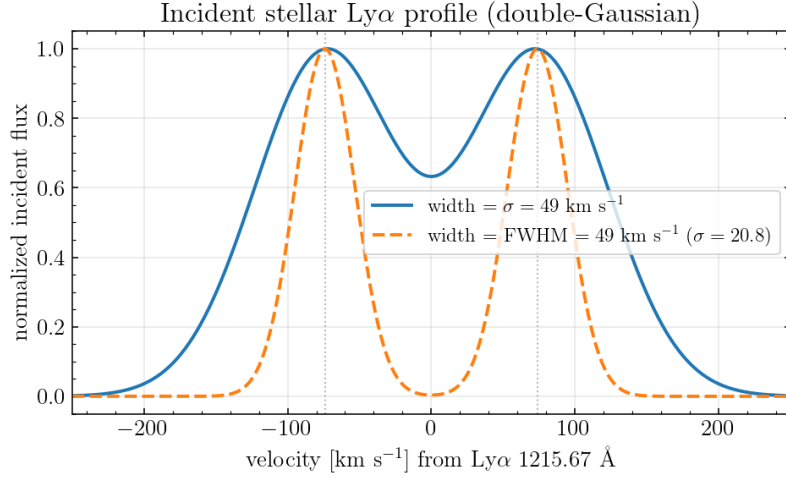


Figure 5: Incident stellar Ly α line profile (Huang et al. 2017 / Yan et al. 2022 double-Gaussian, peaks at ± 74 km/s). The width 49 km/s can be read as the Gaussian σ (blue; broad, shallow central dip) or as the FWHM ($\sigma = 20.8$ km/s; orange, narrow peaks with a deep self-reversal). This shape is sampled by LaRT as the incident spectrum; the total flux is fixed at $F_{\text{Ly}\alpha}$ (Table 5).

the cylindrically-symmetric $P_{\alpha}(\rho, z)$), so it must use the full flux at the orbital distance ($F_{\text{Ly}\alpha} = 37,300 \text{ erg cm}^{-2} \text{ s}^{-1}$). Applying the $\times 0.5$ (or $/4$) redistribution to the Ly α as well would double-count the geometry, since LaRT already carries it. This mirrors Yan et al. (2022), who use the redistributed XUV for the 1-D hydrodynamics but the full incident Ly α for the LaRT transfer. Consequently the $0.3\text{--}0.5\times$ scaling that matches the observed H α should be read as a constraint on the *actual* (unobserved) stellar Ly α flux of WASP-52 being below the EP Eri proxy — *not* as the 1-D redistribution factor; the numerical near-coincidence with 0.5 is not physically the same effect.

Table 7: H α line-center transit depth [%] (negative = absorption) vs. incident Ly α flux scale $F_{\text{Ly}\alpha}/F_{\text{Ly}\alpha}^{\text{Yan}}$. Observed ~ 0.7 to 1%.

Domain	1.0	0.5	0.3	0.2	0.1
$10R_p$	1.46	1.06	0.85	0.71	0.52
L_1 (Roche)	1.28	0.93	0.72	0.58	0.37

Matched spectrum. Taking the binned line-center depth of $\sim 0.77\%$ as the target, the L_1 (Roche) model matches the observation at $F_{\text{Ly}\alpha} \approx 0.35\times$ the EP Eri-proxy value for the $\sigma = 49$ profile (giving 0.78%), and at $\approx 0.25\times$ for the deeper FWHM=49 profile (0.78%); the two scaled profiles are nearly identical (Fig. 8), differing only in the flux factor needed because the FWHM interpretation pumps slightly more strongly. Either way the scaled profile passes through the observed H α bump, i.e. the line is reproduced if WASP-52’s intrinsic Ly α flux is $\approx 1/3\text{--}1/4$ of the adopted proxy — the H α thus serves as a Ly α -flux diagnostic rather than an independent test of the escape model, with the line-profile interpretation absorbed into the inferred flux.

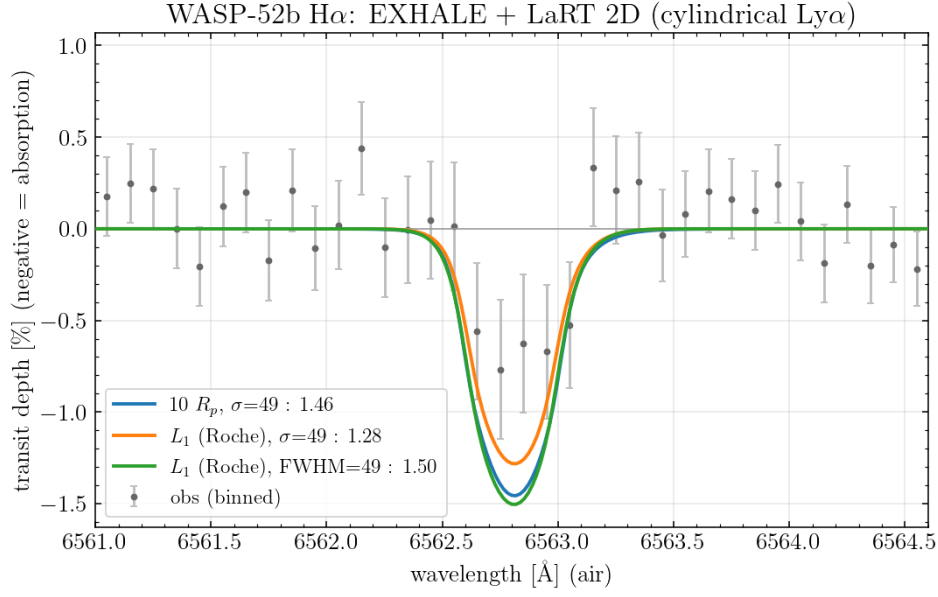


Figure 6: LaRT-coupled H α transit transmission (EXHALE + LaRT cylindrical Ly α , 2-D $n_{2p}(\rho, z)$): the $10R_p$ and L_1 -truncated domains, plus the L_1 run with the Ly α width read as FWHM (vs. σ), overplotted on the observed binned transit spectrum (transit depth, negative = absorption). H/He=98/2, $F_{\text{XUV}} = 0.5 F_0$, $\beta_m = 0.218$.

9. Conclusions

1. EXHALE reproduces the Yan et al. (2022) steady-state mass-loss rates at every XUV flux to within $\sim 10\%$ (Table 2).
2. The He 10830 line depth is controlled by the metastable He 2^3S population. An extended $10R_p$ domain overpredicts the absorption by 3–10 $\times$ because the outer He 2^3S tail beyond L_1 dominates the impact-parameter integral; the bound atmosphere ($r < L_1$) alone gives the observed depth.
3. With the L_1 -truncated domain, a He-poor composition (H/He $\simeq 98/2$) and the best-fit flux ($0.5 F_0$), the model reproduces the observed He 10830 line ($\sim 3.9\%$ vs. $\sim 3.4\%$), independently confirming Yan’s requirement of a high H/He ratio.
4. H α is computed with a proper Ly α radiative transfer (LaRT, spherical illumination) coupled to a 2-D $n_{2p}(\rho, z)$ population. At the He-fit composition (H/He=98/2, $0.5 F_0$, $\beta_m = 0.218$) the model gives a line-center depth of 1.5 to 1.3% (vs. observed ~ 0.7 to 1%) and, unlike He 10830, is essentially insensitive to the outer boundary because the n_{2p} layer is concentrated at $r \lesssim 2R_p$.
5. Being Ly α -pump-dominated, H α scales with the incident Ly α flux; the observed depth is matched at $F_{\text{Ly}\alpha} \approx 0.3\text{--}0.5 \times$ the EP Eri proxy value adopted by Yan (L_1 model). The $n = 2$ Balmer-continuum photoionization (Γ_2) is negligible for this cool K-type host and does not lower H α .

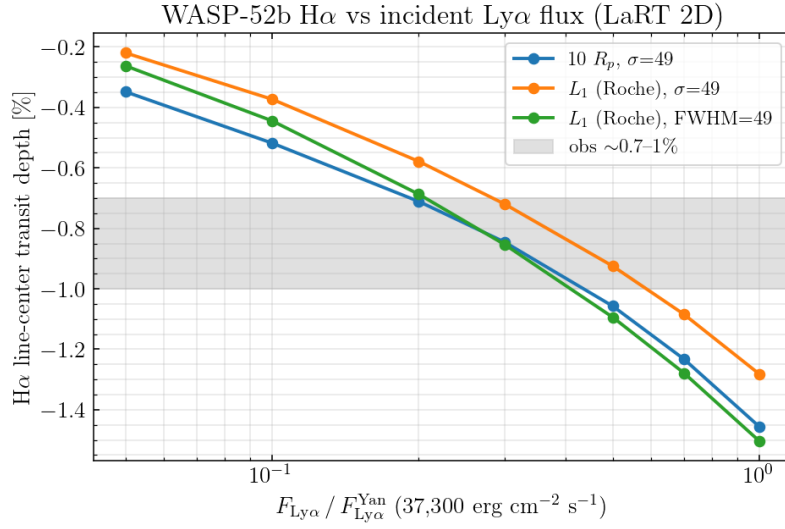


Figure 7: H α line-center transit depth (negative = absorption) vs. the incident Ly α flux scale (LaRT 2-D, $L_{\text{Ly}\alpha}$ rescaled only). The shaded band is the observed depth (1 to 0.7%); both domains cross it below the EP Eri-proxy value.

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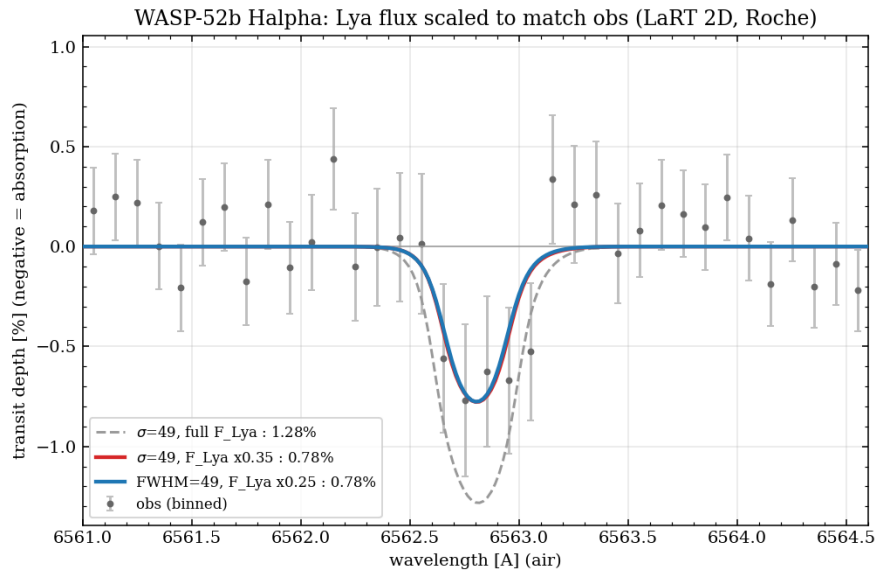


Figure 8: H α with the incident Ly α flux scaled to match the data (LaRT 2-D, L_1 /Roche). Red: $\sigma = 49$ profile at $F_{\text{Ly}\alpha} \times 0.35$ (0.78%); blue: FWHM= 49 profile at $\times 0.25$ (0.78%); both match the observed $\sim 0.77\%$ bump and nearly coincide. Dashed grey: $\sigma = 49$ at full proxy flux (1.28%). Only $L_{\text{Ly}\alpha}$ is rescaled; the atmospheric structure and LaRT $P_\alpha(\rho, z)$ are unchanged.